

Atharv Vatsal

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Experience

Intern – Department of Digital Technologies & Governance, Government of Himachal Pradesh

May, 2025 - July, 2025

- Applied machine learning and data science techniques to support digital governance initiatives and public data management systems.
- Performed data cleaning, preprocessing, and exploratory data analysis (EDA) on large-scale administrative datasets, uncovering actionable operational insights.
- Developed and evaluated predictive models using Python, Pandas, and Scikit-learn for trend forecasting and data-driven resource allocation.
- Designed and deployed interactive dashboards and data visualizations using Power BI and Plotly, improving accessibility of insights for non-technical stakeholders.
- Implemented computer vision solutions for automated document processing and image-based data validation, enhancing efficiency and accuracy of data workflows.
- Prepared and presented data-driven reports to senior government officials, contributing to strategic decision-making and digital transformation efforts.

Education

Vellore Institute of Technology, Vellore, Tamil Nadu

Expected Graduation: 2027

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, Computer Networks, DBMS, Cryptography and Network Security, Software Engineering, Computer Organization and Architecture, DSA, Embedded System, Web Programming, Compiler Design, OOPs, Calculus, Differential Equations, Complex Mathematics, Probability and Statistics, Discrete Mathematics

Projects

keeper.raw | AI-Powered Offline Photo Culling Engine (In Development) ([Github](#))

- Building a cross-platform edge AI application to automate photo selection from large RAW datasets (3,000+ images → minutes)
- Architected a multi-stage CV pipeline (YOLOv8 ONNX + MediaPipe + Laplacian-based focus scoring) for sharpness and blink detection
- Designed scene clustering system using perceptual hashing + timestamp segmentation for burst grouping
- Optimized CPU-only inference using quantized ONNX models, eliminating cloud dependency
- Integrated XMP sidecar export, enabling seamless workflows with Lightroom/Capture One

LZ-Quant | Real-Time AI Trading Engine ([Live Demo](#) | [Github](#))

- Developed a **low-latency trading system** integrating **DistilBERT (LoRA fine-tuned) sentiment analysis** with automated execution logic
- Achieved **~5ms GPU / ~55ms CPU inference latency** via ONNX optimization
- Built **real-time data pipelines** ingesting Reddit, RSS, and WebSocket market streams (Binance, Alpaca)
- Designed **Z-score-based divergence strategy** with risk controls (stop-loss, trailing stop, max drawdown)
- Implemented **full-stack monitoring dashboard (React + FastAPI)** with live P&L, latency, and signal tracking

HP Police ReportStream | AI Document Processing System ([Github](#))

- Built a **production-grade desktop application** for automated extraction of structured data from unstructured police reports
- Reduced manual processing time from **15–20 minutes → seconds**
- Designed **hybrid NLP pipeline** (Regex + spaCy NER + BERT + LLM) achieving **~95% accuracy**
- Implemented **cross-field validation and 10K+ rule-based typo correction system** for high reliability
- Packaged and deployed as a **zero-setup Windows application** for non-technical users

PhysicsRL | Domain-Randomized Reinforcement Learning Framework

- Developed RL agents using **PPO and SAC** for continuous control tasks (locomotion, landing, balancing)
- Implemented **domain randomization (gravity, friction, mass, motor strength)** for robust policy generalization
- Designed custom **reward functions and MuJoCo environments** for high-dimensional control tasks
- Achieved **zero-shot generalization across unseen physics conditions** (e.g., Mars gravity, low-friction surfaces)
- Exported trained models to **ONNX for browser-based deployment with WebAssembly simulation**

DriveSense | Autonomous Driving Perception System ([Demo](#) | [Github](#))

- Built a **real-time perception pipeline** combining **YOLOv8 (object detection)** and **U-Net (semantic segmentation)**
- Trained on **70K+ images**, achieving **55.6% mAP** and **39.8% mean IoU**
- Engineered **low-latency inference (~14 FPS, ~70ms pipeline)** on consumer GPU (RTX 3050)
- Developed **batch and video processing pipelines** for large-scale evaluation (20K+ images)
- Designed **HUD-based visualization system** for real-time interpretability and diagnostics

Skills

- Languages:** Python, C, C++, Java, SQL, JavaScript/TypeScript, Rust, Bash, MATLAB, GO, R, Linux Scripting
- AI/ML & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, NLP, LLMs, Computer Vision, Reinforcement Learning (PPO, DQN, SAC), Feature Engineering, Model Optimization, Model Evaluation
- Backend & Systems:** FastAPI, Node.js, REST APIs, WebSockets, Real-time Data Pipelines, System Design, Asynchronous Programming
- Computer Vision & AI Tools:** OpenCV, MediaPipe, YOLOv8, ONNX Runtime, CUDA, Image Processing, Video Analytics
- Data Engineering & Analytics:** Pandas, NumPy, Dask, GeoPandas, Data Preprocessing, EDA, Matplotlib, Seaborn, Plotly, Power BI
- MLOps & Deployment:** ONNX, Model Optimization, Inference Pipelines, Edge AI (CPU Inference), Model Export & Deployment
- Tools & Platforms:** Git, GitHub, Docker (basic), Linux, Firebase, Netlify
- Core CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks